

Semantic Audit: A Lean-Verified All-Density Exclusion for CMV Four-Arc Minimizers

Independent semantic review completed and accepted by
openai-codex/gpt-5.6-terra

Semantic-audit verdict

Verdict. The frozen Lean package and the author handoff are equivalent for the modeled claim. The primary root quantifies a modeled four-arc candidate, assumes $1 < \lambda$ and the formal CMV type-iv predicate, and negates the model's universal equal-weighted-area, weighted-perimeter-minimality predicate. It does not normalize every source-defined CMV candidate and does not identify the source reduced-boundary perimeter with the model's complete-frontier perimeter. `[@lean:CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_density_gt_one]`

Scope and evidence

The inventory and handoff have the same 464 declaration names, each occurring once; every inventory entry is non-generated, and no generated family is configured. The SHA-256 values of all 11 inventoried Lean source files match the frozen inventory. The reviewed source contains no `sorry`, `admit`, or declared axiom. Each ledger statement was compared with its exact frozen declaration, including its hypotheses, quantifier structure, domains, endpoints, and conclusion.

Model and geometry fidelity

The model uses the strip density equal to 1 on $|y| \leq 1$ and λ outside. It defines weighted area on the coordinate carrier and weighted perimeter against one-dimensional Hausdorff measure on the complete Euclidean frontier transported to that carrier. A four-arc candidate records $0 < h \leq 1$ and $0 < \alpha < \pi/2$; the CMV predicate separately adds $1 < \lambda$ and $\lambda \cos \alpha = h$. Minimality ranges over all admissible equal-area competitors. `[@lean:StripDensity]` `[@lean:FrontierMeasure]` `[@lean:FourArcCandidate]` `[@lean:FourArcCandidate.SatisfiesCMVTypeIVHypotheses]` `[@lean:FourArcCandidate.IsWeightedPerimeterMinimizer]`

The type-iii assembly is an actual carrier with the major, semicircular, and minor cap branches, including the degenerate bottom segment at $h = 1/2$. Its carrier-area and complete-frontier-perimeter definitions are distinct, and its separate integrability theorems establish the corresponding area and frontier obligations.

`[@lean:TypeThreeAssembly.branch_complete]` `[@lean:TypeThreeAssembly.WeightedArea]` `[@lean:TypeThreeAssembly.WeightedPerimeter]` `[@lean:TypeThreeAssembly.integrableOn_carrier]` `[@lean:TypeThreeAssembly.integrable_frontierMeasure]`

Certificate coverage

The source establishes a stationary type-iv fold in $(9/10, 1)$, constructs a stationary equal-area type-iii pair with strict perimeter improvement, and transfers that improvement through the descending equal-area envelope to every modeled $0 < h \leq 1$, including $h = 1$. The geometric bridge turns each scalar witness into an admissible type-iii competitor and thereby refutes modeled minimality. `[@lean:LeanSuffixAnalytic.typeFourFold_zero_exists]`

[@lean:LeanSuffixAnalytic.stationaryEqualAreaPair_strictImprovement_exists]

[@lean:LeanSuffixAnalytic.allCurvature_typeThreeImprovement_of_envelope]

[@lean:CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_analytic_typeThree]

External-source checks

The Cañete–Miranda Jr.–Vittone citation resolves to arXiv:0906.1256; its cited §3.2, Lemma 3.8, Conjecture 3.12, and equations (26)–(27) are the stated strip-density and candidate-classification context. The supplied Lean 4 and mathlib DOIs resolve to the cited publications. No external citation lacks a stable locator.

Disposition

Accept the package as an exact proof of the frozen modeled claim. The result remains conditional only in the stated modeling sense: source-level normalization and reduced-boundary compatibility are not proved or claimed.

Package-level findings

Relation: equivalent

No package-level semantic errors were reported.

Declaration-by-declaration audit

AdmissibleCompetitor

Source hash: 52c3c4597213ca5cc7d4e12b21ad852ff148bcb8179b3f52a9c03b7d3665311b

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3952-3960; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

AdmissibleCompetitor.WeightedArea

Source hash: d58b51a28690771d1b404e9048905bb6eae43f08f6798b27166b1234f3251f5e

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3968-3975; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

AdmissibleCompetitor.WeightedPerimeter

Source hash: 17b5c5914d38bc0479d502c48cde9715c303aa7070585ade87ba56d307321437

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3976-3980; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

AdmissibleCompetitor.carrier

Source hash: b57d4c4c33776a05ffbdbc22f1b363a7f356aa7cd089d7c2b3335ead1bb65644

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3961-3967; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

AdmissibleCompetitor.weightedArea_eq_carrier

Source hash: bf49ad5d7ddb4e090057dafec5dba684f6e36be017290e55451b4a9c6cfabce1

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3981-3990; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.FourArcCandidate.endpoint_weightedArea_eq_typeFourArea

Source hash: 1d3c440482f713868366732790a87ef12add8058ecc71af2e6fb07df04bb3b83

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:159-166; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.FourArcCandidate.endpoint_weightedPerimeter_eq_typeFourPerimeter

Source hash: 9a9144c9e0869ee21b16e09b6aad1d31570ea82a7d5d876263c80c006e5e5661

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:167-176; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_analytic_typeThree

Source hash: 99c923b45cbc09e9ba012e383efbf5a498fe0a194180526cbe54c89d50a0f151

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:201-224; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_density_gt_one

Source hash: 2bcc5fee077e8fcf68df34eb9a91b0e0249285ccf8a646ec28b5a30fbd597c99

Relation: equivalent

Checked against the exact frozen declaration at UniversalStationaryPair.lean:146-158; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_stationaryEqualAreaPair

Source hash: d6670f69ae0dd1a579233b44583ff3b5ab90f13ac210ff1ddaabbb1088c5b5e2

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:225-248; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.FourArcCandidate.regular_weightedArea_eq_typeFourArea

Source hash: 48c010f5756d8d71b74c4c11f50c789619d79d3e32a79f62433bedf4290ea164

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:142-149; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.FourArcCandidate.regular_weightedPerimeter_eq_typeFourPerimeter

Source hash: 7c7abd9d4131a70966d7ee06757ae9000eb6c0370ad6826014e6939680589328

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:150-158; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.FourArcCandidate.weightedArea_eq_typeFourArea

Source hash: e8473928799276d8ba026ded04a5483fef2171ee9679b0cfa48b5117289324b3

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:177-186; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.FourArcCandidate.weightedPerimeter_eq_typeFourPerimeter

Source hash: c47649486a6e39478b2697659276a1423a248517e94449da31be1eb9b63be5ce

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:187-200; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.TypeThreeAssembly.angleTerm_eq_typeThreeAngle

Source hash: fe9ab761a6d7819237c18cc5b6d5b714cc699cb312b84007d27510fce813ca50

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:42-52; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.TypeThreeAssembly.scalarWeightedArea_eq_typeThreeArea

Source hash: 09eff0e6af5f2cf4916a82b585443109aad4f3e8bb020859b5a5f6526f9c2f0d

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:78-86; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.TypeThreeAssembly.scalarWeightedPerimeter_eq_typeThreePerimeter

Source hash: 1cbaa2764b23e5bf11163fb498e9653b454e16f1f8f6b84d752362e7a842718b

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:87-99; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CMVSuffixModel.TypeThreeAssembly.sineGap_eq_typeThreeDelta

Source hash: 0b54dd8dd926b7670949326ff7f7fb7d21250d76068f1da2eb207447216f1aac

Relation: equivalent

Checked against the exact frozen declaration at CMVSuffixModel.lean:53-77; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

CapSide

Source hash: 6863ffc025769eb8169e9582a3d673fe180025902f766c2557257f6937cc7a7e

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:92-99; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

EuclideanPlane

Source hash: dfa2db26da624a2b5333b886cff8c9cd3151218ea8aa3e3cb39c78359b32470e

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:34-37; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FinitePerimeterRegion

Source hash: 2efb7176d17b946809f4cd07b78237ddd03cd4197c405c38d08e97342106a1a0

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:74-82; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FinitePerimeterRegion.weightedArea

Source hash: 3892310b68bd795c4ddff8f1d276f58be0efcb96c1e8c4fad96ec7d73deea20a

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:83-85; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly

Source hash: 1ac9185bddaf1e6a0074fb39d1e29417262cc431ccce861d3c65ec2e5216fdb1

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2491-2498; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.boundaryMeasure

Source hash: a07e8d5b98aab97a737c50d9ee9eae871142e1adb2b58a3132be8c4b6a093cce

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2525-2529; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.boundaryTrace

Source hash: d8944f364e38eaa5c7b77ce182d417bec91ac425efc229739d886be00ad658c6

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2625-2630; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.boundaryTrace_subset_frontier_carrier

Source hash: 160385ee99b171abc63b7cc7cb5ade687af8d5249c1ab26a5150be49b8c4654b

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3244-3251; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.carrier

Source hash: a55616796db22264c4fe11d48e20ab173ecb661c3bdced26752617ea90d1f816

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2519-2524; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.frontier_carrier

Source hash: 7da383d241cb3cc30fc9da784ef7200d9bb87d82e25006745acf6f7f7d86a78f

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3252-3259; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.frontier_carrier_subset_boundaryTrace

Source hash: 0d72e7b21e3528564863313aa2366a2e4806e01c00a4772872911d424abf3cc5

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2795-2819; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.lowerCap

Source hash: a60a9393b363872f08f12dbab41380b51afefa746b9504c5841f28d389091bfe

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2509-2518; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.upperCap

Source hash: 3f084f0dbec6bee8992e556fe2ca304aae5eb8981815636bf15454a3b8e613f9

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2499-2508; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.weightedArea

Source hash: ad43970e4ac79b935baed41701092f4d901053e35bb88583dcbd5a9003db13a3

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2541-2545; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.weightedArea_formula

Source hash: 839e3c1d7a83b5801ac118ef77feb57b415f9e5a71dfa154b8987ba3b91118a0

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2554-2602; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.weightedPerimeter

Source hash: df8ef218db7076c4681d4506265ae824cfa4b1a2ec872c320aeb4f283b25493b

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2546-2548; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcAssembly.weightedPerimeter_formula

Source hash: 43ff4cdbc6774032399c0e7d7f5767a18500986a522e0ace1e470e2035290acd

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2603-2624; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate

Source hash: 24dc0ff22e0918bdb2c7f16fab0da52bc08326d88e5cb8e1d2e8a8f7ec2bee62

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:23-35; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.IsWeightedPerimeterMinimizer

Source hash: f04b5c158c1a510cd5a8af38df8f99424aead5e376f986e1d437f95e6a4cbe17

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:90-94; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.SatisfiesCMVTypeIVHypotheses

Source hash: c09b9c7180b6131040333c0b3c2623f3c89af7f54fe8ab0b9f6a41be873ca0dc

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:84-89; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.WeightedArea

Source hash: 8256dd516e32d853aaed6a85d09fdb0b62411bd0bf67f1df7ffd123e47baed99

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:76-78; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.WeightedPerimeter

Source hash: d649676f2fd717389a6e9a17a385fe854bc399e17ecfdad24bd8a7bcc9a36521

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:79-83; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.alpha_eq_arccos

Source hash: 46219b8ad0fc6b679968f8972242a6ad511174ce969ce84539a4054206de6595

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:276-292; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.assembly

Source hash: ca67fddb2d167afab63544438387902bb645349ebfde9d83c7a2719a2277c941

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:66-72; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.beta

Source hash: f3520601e601026585d5024a59c3bee3be514d265897b65e66919a9d3ea7fd1e

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:39-41; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.candidate_area_formula

Source hash: b6d944ecc5d4bab77cadf263520f194142a9d6855088a9b98e5873be144ab8cc

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:410-430; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.candidate_perimeter_formula

Source hash: b4949bdc7019f429151f2faf734db8466461943a33be006f4c104caebb493fe0

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:391-409; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.capArea

Source hash: eb6c25ab8d5e1e4d2ce589b98fc3af2d04e58bd682ad273d1b07834426ee23d0

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:45-49; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.capArea_eq_chord_sq_mul_area

Source hash: 0e9545ae1662d69d55500dc9264e42c4c5b03b78571f6473a85822aca53c6073

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:104-109; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.capChord

Source hash: c09aa141e8bbabbe7279bce2473dd47aa931f875442fe1fc79d759685536fb1e

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:42-44; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.lowerCap_euclideanArea

Source hash: 14a860cf99daaf759c3fa7838aed86edfb2879de1a5a3f6f7d84f9996c7da155

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:159-166; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.not_isWeightedPerimeterMinimizer_of_typeThree

Source hash: 6807359794d5b2629c85ed715f93a0b06f1e0f5dc3b504c99a54401226562f55

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:3101-3118; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.region

Source hash: 8ab9f84c197a1a6dbe90236469e9ff5cc2696f04e42133db44cdf34546558b75

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:73-75; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.side_caps_area_formula

Source hash: a639bd8c9eef40f2c4f87fdb62b671cf7db0ce41013e7d923721f6f472bc8cf1

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:351-363; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.sin_alpha_pos

Source hash: b92e14d3743fcaa1d6c447d96c6334ba3f5f129152aa26ff47b97d6fd8de2e28

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:95-98; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.stripCore

Source hash: 66aff9e1eb934bbb8dd09a1a68a4c23aef3ee09735d3ba9bc12eadb4664ee

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:53-65; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.upperCap_euclideanArea

Source hash: 63eb34459de3e09e8c932883edd9446c50efc2cc1a148ef60c5c77cbd56d7496

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:153-158; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.weightedArea_components

Source hash: 654554fba4e2f4847eae4f853550061afa29cf158da02de52109cfffacedfd702e

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:364-373; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FourArcCandidate.weightedPerimeter_components

Source hash: d02953cec6a51d470b1b940fa2318bbd5c86354d0e420d179322c86356c9434a

Relation: equivalent

Checked against the exact frozen declaration at FourArcCandidate.lean:374-390; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

FrontierMeasure

Source hash: 7323bb80c0a565daf410d27d199e2143ee33a0ecca0460c52504f2e7e50b48e2

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:61-66; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

HorizontalSegment

Source hash: e7bebc91d2e392d80b21e5e8b5487591a5477fe10be16627e5b6180dd120506

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:392-399; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

HorizontalSegment.carrier

Source hash: 094b96b3006204081e6d336250f56e63f6b337cc71255210b9bae8ae3b894041

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:406-408; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

HorizontalSegment.euclideanLength

Source hash: c965c7e2ff7089171e9318c88e3733eb9629da0051af6a9aad082bf1ad5a5aef

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:409-415; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

HorizontalSegment.leftEndpoint

Source hash: d64f9e0ea48cfbbc1b06bed874068659f0011957eca2b8226c738430113a0615

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:400-402; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

HorizontalSegment.rightEndpoint

Source hash: a14791633639531ab0cb8801c8e0e16383dd7712b7d5730aa7f0d9178c83e5da

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:403-405; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.DescendingEqualAreaEnvelope

Source hash: ee2db888cf8e5e37251e350d2425fb86808e5c1f7d808208572e20c0e3ea40a9

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:279-300; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.FoldGapNegative

Source hash: e0a51358d0910c8fb2dd4ffa9d95125b8210328ea93cd0771ca9e9c1138b2ced

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:577-581; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.StationaryEqualAreaPair

Source hash: 865710967f267b9ec6707c5d925cd799ff4862bc66ec64cf0a22afce00862eef

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:1177-1188; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.allCurvature_typeThreeImprovement_of_envelope

Source hash: c4f7d83c3e3a4e728eb9be863cb4e20b5d04c133005f9a44bb6548808441f1ae

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:645-658; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.centered_meanValue_exists

Source hash: 9e16c64246998ba49ad719b4074913f0337c8612efed42e98f32935fcd55a3e5

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:93-129; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.equalAreaPerimeterGap

Source hash: e8643a1549a791c31f1d0a91f7cb9b302d6832ce9c0d1d8e354d9c3e22351fcb

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:34-39; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.equalAreaPerimeterGap_hasDerivAt

Source hash: 42f42f4f683c9b8ff6e876958c9ae3b87d6bfd1ebf63072c7e612401d7b5ee93

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:184-225; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.equalAreaPerimeterGap_le_fold

Source hash: 0bde810fe503972e30290f54b6517ae65288b7e84131b161bebac58ed0796e33

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:632-644; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.equalAreaPerimeterGap_lt_fold

Source hash: 1a0b26ad0e4af7fdd933118930486f792fc517471c75bd1495b190e7f97f3420

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:582-631; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.equalAreaRoot_continuousOn_of_compact_unique

Source hash: 58b73f54ea82a27e264112e64332402373c869c542bc7277160adca9dcc99fdd

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:71-107; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.equalAreaRoot_hasDerivAt

Source hash: 4240cf112f1c537233001e48d7956d1d7de811e741deeb8b94332f375fdbb1f0

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:165-183; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.equalArea_gap_identity

Source hash: 7a449a0d73201ece195743b096dbde5032c8598ab1748985129132f301a8ba9d

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:1208-1220; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeFourAngle

Source hash: ed3b0f3804a218b0cd68be6333d68c33832835819c8ca12dd83eb68bae07439d

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:241-278; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeFourArea

Source hash: 406477265a826e873557d3950f02d636becf1dd68c773947e69154fec9631e9e

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:496-553; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeFourDelta

Source hash: 71475be52cec7e059c20046cff749d550fce2a4275677924c383af205dd192c8

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:279-305; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeFourFold

Source hash: df765bf2b71951f20ce86d7897c71befd43183478a79e7706b6061a82ade97f0

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:457-495; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeAngle

Source hash: 6ec918819525fbc766cbbf40969985e480000f0bcbcf23d536b033e9e80b6380

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:154-205; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeArea

Source hash: 7e0c0da90901c546a8edb152b4ed9a40769e6bd51ef8318f50390135bcace455

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:700-724; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeDelta

Source hash: 6a259e36542b424c2657b96b59d66f9d38d054281073fe5c590c3110a33522c8

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:206-240; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeFold

Source hash: c8c79d60a2ba5a7f202a48aa5b5851e2b132f448f609c9f5925a2a8edab62ee5

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:375-456; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeShape

Source hash: fc5b59d714869fde00b69f1950c83bd7a32c47524e01ad8d04455ff346c39d58

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:146-153; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.radicalCubeGap_pos

Source hash: ed04fbfe0d2a5c8c68ae2429d8da8593a273f2471a8fcc329e85c97a6780176d

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:349-374; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.reducedFoldGap

Source hash: 90c9f564cc40a2e05814e95c52b8ea113ecc45335e828ff2a54c8e9f2caf08b3

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:83-92; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.sameHeight_support_identity

Source hash: dfaacd20b555a53066a164daf8f9b4ebb1dbcc86f6d564ed5188ef405330e787

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:348-361; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.stationaryEqualAreaPair_exists

Source hash: 2bd42d07c65a61210061f6c5531a6ba1c7c3d14d12463bb14fd552e96e5e36b6

Relation: equivalent

Checked against the exact frozen declaration at UniversalStationaryPair.lean:107-115; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.stationaryEqualAreaPair_of_typeFourFold_zero

Source hash: 87c6601c0e4b1d865832811725296f245fb9e4dc0b97693804d443361bf40752

Relation: equivalent

Checked against the exact frozen declaration at UniversalStationaryPair.lean:62-106; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.stationaryEqualAreaPair_strictImprovement_exists

Source hash: 1134a3b056794ea4a2511e4b1ea810f5acd87b2a3db5089871809393e30f50ab

Relation: equivalent

Checked against the exact frozen declaration at UniversalStationaryPair.lean:116-145; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.stationaryEqualArea_gap_identity

Source hash: 7ca190ef8d3e84eb702882d4666162a0576db626556bb2591b0ba437f522132f

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:1221-1238; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.stationaryEqualArea_typeThree_improves_iff

Source hash: d313a9a9b483a140c805abe01de0a03576005db4ae69b2c417e95c5e91300b63

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:1239-1252; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourAngle

Source hash: 94e6a9f5b491cb7ca4802d89fba1fa7ccb864bbe777d6e41d89b1d6bf892bdd0

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:53-56; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourArea

Source hash: 775f7921b66716563dd5e0e4c35562d880f5cef2a71b40cef616cd10eb9488d2

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:61-64; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourArea_deriv_neg_before_zero

Source hash: e504f043a30eb739ae5f3ba9174d0ca2cd6aa51665b8ee6815ad3b1b3ea9932b

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:1071-1082; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourArea_deriv_pos_after_zero

Source hash: 574cf814295ee9aa4cb9406582955f50506dd66296979a4f903f27b9ae8e2667

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:1083-1094; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourArea_fold_le

Source hash: 00808b809e3e67ecf06ff39be8e712a367908c043a9c526b98df7d97ff33901d

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:1095-1129; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourArea_one_lt_typeThreeArea_half

Source hash: 10ba0ee6421be27f2e3c7dd8f7b1b5dc586c66d90118dd94301a5307188face6

Relation: equivalent

Checked against the exact frozen declaration at UniversalStationaryPair.lean:38-61; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourClearedFold

Source hash: 1266cefaf66b4dab37ed6ce0451a3ec8c8e11c97af24ed298db2b491c2b5e620

Relation: equivalent

Checked against the exact frozen declaration at TypeFourStationaryRoot.lean:24-28; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourClearedFold_continuous

Source hash: c2810ce70b601a04bf730b7d74e237dfa7c31c5a16c6857177a5231549ec1ba2

Relation: equivalent

Checked against the exact frozen declaration at TypeFourStationaryRoot.lean:29-34; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourClearedFold_continuousOn

Source hash: aad0d8aa37abd7a9983d9c3fef33a66d571f8cc007154660095f3a1d473e9d30

Relation: equivalent

Checked against the exact frozen declaration at TypeFourStationaryRoot.lean:35-40; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourClearedFold_eq_fold

Source hash: 45ea307682f4ec267ee71b675163dd61e82ecf847d6e7f8bf8dbc13d2cb15e17

Relation: equivalent

Checked against the exact frozen declaration at TypeFourStationaryRoot.lean:41-54; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourClearedFold_multiplier_pos

Source hash: da4a3633659e62178714586a00f2c1bd42b925fbeatbc151bdee7d46c4c36171

Relation: equivalent

Checked against the exact frozen declaration at TypeFourStationaryRoot.lean:55-61; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourClearedFold_nine_tenths_neg

Source hash: 08d09dc1047004d87702d12abfa5bdbdcddf35bfb88c89bf559151153a6e642e

Relation: equivalent

Checked against the exact frozen declaration at TypeFourStationaryRoot.lean:112-124; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourClearedFold_one_pos

Source hash: 5b0cd6de2bd6df7214891200ea0e61cc8d389c1b1edf8e7a99120d747111a4bc

Relation: equivalent

Checked against the exact frozen declaration at TypeFourStationaryRoot.lean:125-131; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourDelta

Source hash: a4c8b30971f59f8f04626dbddf49af92c98dbfa9126bf3766c0be8b6472cbfb3

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:57-60; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourFold

Source hash: e492724315b735482cab607f768694fcd028864cb336800b7469d841099293ee

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:78-82; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourFold_strictMonoOn

Source hash: 16af12c38f3bb1c629d7204f781995879e693ae4c65a0b5d7c3475b2043d0222

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:1056-1070; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourFold_zero_exists

Source hash: d603264a772d81bf55a0dad7fb9b6a0704e412d433e0051ac8f104d2554f1f15

Relation: equivalent

Checked against the exact frozen declaration at TypeFourStationaryRoot.lean:132-167; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourPerimeter

Source hash: ad0bebbd79c17351c0f12ce6a7fdae3e930dba33de0077415a6116df1d1e6e25

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:65-70; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourSupport

Source hash: fdeb0489b7e6f5e26dfa807a57579da3ecea22f44d0bba90cd5fa78c0744aa33

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:266-269; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourSupport_concaveOn

Source hash: 8ef0a051daef9d1f2bbdf8d2a9a39d0e0a16c808a777adfdd62560b57626cb9b

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:293-311; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourSupport_hasDerivAt

Source hash: 7ec9ad85e9c92ae246dffaf1ad5a0c048a08bd6754c8a63a3be2760ea37a093f

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:270-292; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourSupport_midpoint

Source hash: fc169235755eca8de1ba9408de035c749430eb621ef56b64e6fdcdc4766386f

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:321-347; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFourSupport_one_gt_halfPi

Source hash: 0c5ddb44fb627a6e8b09e47012b025857f4fe02fa3598a05502fda2e5e2e8921

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:312-320; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeFour_variational_hasDerivAt

Source hash: be216789e3804b1c530f202d3470319a386e5c80dc03952cfaf3ebc685f27297

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:733-753; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeAngle

Source hash: fb7617b8b745792c6f019f3cbcf3c26d5e54f1fc7dd47bb69cec919c712d110a

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:35-38; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeAngle_add_halfPi

Source hash: 309f9c361ad038b7eae75200b2b081130c71a155a43f2f07c9ad2ae282ce68ff

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:130-138; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea

Source hash: 63d13d5879252e16ac6f53a2d4449b4f511e353d6016e98158793484dcfaa8f7

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:44-48; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeAreaSupport

Source hash: 59983bd94c5b2c86b0a792e561e5d858c16e3c1a5d5652441772674daa8dcfc8

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:464-466; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeAreaSupport_hasDerivAt

Source hash: 84f5f511ccfd2c66897b9ce22e970b7b5810edf0c56f08f711aad8e54224205e

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:467-480; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_centered_meanValue

Source hash: 0495f3e7854574a7754a0b63ac7fd008229aa9ba79f907c1f5c180c5d5dea354

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:778-789; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_continuousAt_one

Source hash: b82ff69676ab90ec4fb13d857723ab9b2dd80ce0482e1e3d55da8f2320d7eef0

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:384-393; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_deriv_eq_explicit

Source hash: ec96d74eb7c231919df47f4c5a7bcce391a0bf68733b2ba3385beb5c882b9ca6

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:649-699; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_descending_root_existsUnique

Source hash: 15660b04d9615f11056408ab4ae99cb03c7fde3d8d26f5056c89e742987b7edb

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:1013-1039; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_descending_root_existsUnique_of_stationary_pair

Source hash: f368bc947db4d0b119fefc4c63c0de6d0d14cb14db7e5a28faa26c535eb880b8

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:1130-1149; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_descending_root_lt

Source hash: f0218743e3c66d2b3ba1e5bdde4d44d6f7c58ed79dc9416fb8fb18d852a14824

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:586-622; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_half

Source hash: 315087e4cfd2ae0e85b582525152b7d3311b9e9d14e2dc9c72058614a85c134f

Relation: equivalent

Checked against the exact frozen declaration at UniversalStationaryPair.lean:27-37; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_le_max_endpoints

Source hash: f271402f8f772a214053257391493497c3abc162da11014d954aaf909e4fe1d5

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:394-463; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_lt_typeFourArea

Source hash: 292616e61180882b74b7405ba97fd168e8ed5d3653115a72ad0fa3e78e338f65

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:225-265; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_root_le_of_fold_neg

Source hash: a9a71e3bf8960b5c96a8fc59bf3fe2a372f70aa365bd91817e7ca128cb8520b9

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:954-973; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_strictAntiOn_Icc

Source hash: fd210f8a463759cd45da345eae2826fb3a7bdf59d29239656d68c8efc64fb7fc

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:108-123; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_strictAntiOn_Icc_of_fold_neg_right

Source hash: 1b1d7870499bdaf74cbcd036814fc1cce139b4cf2463cce4ba39b71ae55024b7

Relation: equivalent

Checked against the exact frozen declaration at EqualAreaEnvelope.lean:124-138; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_tendsto_atRight_zero

Source hash: e82ffa97f750ea7c64d3d22eebba58e03fe2fc0f731297b9787ae711ded5a49a

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:828-871; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeDelta

Source hash: f96722ed089f4e5911c6013a0aa601eb4080090036491702cd0fa78e3e31113f

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:39-43; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeFold

Source hash: de466d8508a0def3484419e5a98402f15940017ff9d116432fb67c21c1d22c86

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:71-77; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeFold_neg_of_equalArea_of_lt

Source hash: b45c2e12676c7119b05c3219fa4f0e77bf807a7780d13de5a358f0b41a3fa64d

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:554-585; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeFold_strictMonoOn

Source hash: 548db5145f9bcf2011e54d3648e735633b97c51f2514a21b4368c3304bb06a32

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:872-893; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeFold_zero_between_of_area_eq

Source hash: 314dbeb5db22bb018c3bb2b871eb2f76a4711df985ead7c117e1cc5036bed173

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:905-925; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreePerimeter

Source hash: 22e35e36f9c0930c9f9233f0004c686c48f6f282298a578def5a6e6bba11f34e

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:49-52; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreePerimeter_deriv_eq_height_mul_area_deriv

Source hash: 7607f8124dad06702d520d2b8e20b8eab5117f85a2007d935f98d72479aad4e2

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:640-648; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreePerimeter_lt_typeFourPerimeter_of_equalArea

Source hash: e7b6d671984a991fd2b92d8c073a4e4e44f40679b3766f18a0ac3843895845e6

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:495-553; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThreeShape

Source hash: 9ef91461acc972465b98decfe31d601788f1354eb0ffbb8d094be6f5c46ca593

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:32-34; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThree_support_lt_typeFour_support

Source hash: 3b4307e15a93ff15bacde69d80f20dc52b17ba9ab4d81965f114554585e3a373

Relation: equivalent

Checked against the exact frozen declaration at SameCurvatureArea.lean:362-383; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

LeanSuffixAnalytic.typeThree_variational_hasDerivAt

Source hash: 727f5f60431aed129e76617edde1e3ccdf8d18a1ae71fbf80ca33147d09b9cfa

Relation: equivalent

Checked against the exact frozen declaration at LeanSuffixAnalytic.lean:554-639; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap

Source hash: a9dd2ba7ee653be367b6a2072de1384c3362b0214117e3aecceae763bc6f8403

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:100-112; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.arcLength

Source hash: 60daa6fa20eb20410aecf4ceebd5a41e50c48a0eb958e6b55f1bcb20cd85ff25

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:167-170; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.arcLength_pos

Source hash: 615464c0ac73967d872be72b9e5d931fb0644d2db70b6e9168ad1dd7c81aa184

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:220-222; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.arcPoint

Source hash: 2f12b44987266410d1b21333c27225fca467817287b4028245b465856f5540c3

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:124-133; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.arcPoint_in_closed_side

Source hash: 4692c99be4beede3a58f1e164acd1cfe21db7a9160da63340de9490acd46c184

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:240-252; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.arcPoint_in_open_side

Source hash: 3dbf6a865d1d6331f9c3406c05af249acf00d7c487349d982b588394a814d697

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:253-266; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.arcPoint_left

Source hash: 4e22ae8901c6f895322d9f1591e370b613671c823882adf707a5fac1ae750287

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:228-232; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.arcPoint_right

Source hash: ba639553cc75debbcac47cb7f164b0cdee0e14e9586cdc6d1cf857f7fa27f21b

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:233-239; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.arcTrace

Source hash: 5e3edb81f4f16ef8dbff7c15d5756a176b1f99c3bd5fb86ce703134a0a727a89

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:656-661; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.carrier

Source hash: e0644dd0b5f330958b79efbb3810eb1cc2f4356e5db251a95736b532b75bfcf1

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:146-152; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.center

Source hash: 58a018a3af71f1eb6c5aebb7ee395b60aab79f7b55a9d384d43694baace0d346

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:117-123; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.chordCarrier

Source hash: 0705736836c9203b120221cb0770c365bada15ab749e72d0791d55f58eba4c1e

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:160-163; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.chordCarrier_subset_carrier

Source hash: a35dcecd8fd218da748131deaec9b02f5328c1034952ad269b6ba1f5b686d947

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:311-334; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.el1_theta_pos

Source hash: 88c4230a8ce8dbe3ed5e64385c18de0588039e6c242ead1e2df6299d087f0ad8

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:216-219; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.euclideanArea

Source hash: 5c476b88f35c9e9f6afc55ed7ed9ef6be1395a7b5a3003452ff8691443c49da

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:164-166; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.euclideanArea_pos

Source hash: 3902ccdc9eb3694014c5a47aad3179ebe09a3fd3cddb066b9c4c1ca933463218

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:213-215; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.frontier_carrier

Source hash: d9cb2c749721b51457faae17a95033cf908527d3fdc652ebedac fbb2292d579d

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:697-761; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.interiorCarrier

Source hash: d9986ab2dd520dfd41d6ccb99ca9878df085185d6c10c136685b43549b4c4dc1

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:153-159; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.interiorCarrier_subset_interior

Source hash: e055c7441e3e2230b6bd3c3989fc3043eb631862e9cdc4e7fe3a7963b0242ddc

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:687-696; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.isClosed_carrier

Source hash: d757ff8cbf988a97323a1f1c27dc667eab9b6475f2eb7d1c16beffa82e3009a4

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:364-391; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.isOpen_interiorCarrier

Source hash: ec8c11c26c05148d770698c167cdec d56869de41dc90864a365cf2514de87bc1

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:662-686; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.leftEndpoint

Source hash: a193f337e9021d7e081f4666c9f784092558b2268ec0e9060f11b3d18dfc3c8d

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:134-137; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.measurableSet_carrier

Source hash: e0f551f525fbf0409babb736205ec484e0a0c2ff7039ce55de1e7a7fc5595a27

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:335-363; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.mem_chordCarrier_of_mem_carrier_of_eq_base

Source hash: 424261710f758c71ebe7e57dd13ebd11ffc59cb51b840ff8fc86319890ddae19

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:291-310; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.radius

Source hash: e898cc7066a75a606fdc3b1d1e563939ad569bdb0f66fd289569e58c7f4c44b8

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:113-116; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.radiusSquaredAt

Source hash: a1c0f970b426c7f0e6a013d8d8bbb613fbba3b5c253445dacecdb9b47cb6785a

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:142-145; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.radius_mul_sin

Source hash: 4b800b6287f8bc78c34b40f39fff02ab7d25631ad2fd22f8819095ceda60143e

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:223-227; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.radius_pos

Source hash: 7bcdcf3411946f45cb3c854e76728bf4e1477fc16c0bd8673b5d0f6f904d6adf

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:210-212; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.rightEndpoint

Source hash: e83d4942af5a95dd72bd914bc15f416e046e239fc23caff7a851807f832f26a3

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:138-141; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.sin_theta_pos

Source hash: 8c22795350624813eb7b9aec2fdcda7bfc438d144ebb38b607db3cac77688583

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:207-209; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

OneSidedCircularCap.volume_carrier

Source hash: 0d51f9d06b45af9eac50b2062d616aee51a260200ad5a2e96d2ecbe8e98be85c

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1996-2028; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

PlanePoint

Source hash: b6c33ab7d9e6652b4833188977c0694ee665f31ebd07e572ea7a67ee7a8a0183

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:30-33; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

ReplacementAssembly

Source hash: e0b73790946058b82b8d6a4bdde47ad7560092e26acf3d58cb0c91dc0cfa445f

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3281-3288; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

ReplacementAssembly.carrier

Source hash: 36b80491dc0fecbf358d87c6bfe7194faf4bcad4ac91372874e8af60a30fd23e

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3305-3309; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

ReplacementAssembly.upperCap

Source hash: b30d86506333a29d0fea0a0c213d5ab5e404eb8902dd90c6019d57dad3b8f56f

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3289-3298; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

ReplacementAssembly.weightedArea

Source hash: 30517c93164485b2476e45ea547b19187ad53db44a33af2bf70144ca4c93c662

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:3326-3329; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore

Source hash: a32ba501747725a0a1307cf7bd2abbfffd387c8bc9d810f24cf348c7c04513aad

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:416-425; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.boundaryArcLength

Source hash: 74e634700740d7651d4ed773b1cb3cb71b11c7eedacd0c71b6a84a0e6ec43665

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:457-459; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.carrier

Source hash: 05f714bedd5ca3f09fd1dac168964dc190e4830b8099d994e590458d8167e259

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:449-452; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.carrier_y_bounds

Source hash: dfd3510aaac02f0e3230984bb1d2516c15f954b7d416fa7e30c71ad44efa3ab8

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:482-490; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.cos_sideAngle

Source hash: 5c26f39a6dccf790fede705362af68051955e479d2108518ff90a5d9985148ed

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:478-481; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.euclideanArea

Source hash: d05ac0be663f49bfb5425917ab3d2dcdd3367e8ef25ec467ccc3d1ef4ccf07a7

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:453-456; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.frontier_carrier

Source hash: 34bdf246812f32ba3d6167bf3b3ff5fea490b0816170740d97a7d77e703585bf

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1088-1199; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.isClosed_carrier

Source hash: d28fca8c993d5d2b9aa0b2ebe7f3f0372ebdd951a5cda90d26adf748bface0e4

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:805-843; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.leftArcTrace

Source hash: 702872b40533619867ab566579d52d3f6a44e812910796d96106d1c50acbbfc2

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:762-766; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.leftCapCarrier

Source hash: 2ca301cf56be587995c60fad814a6021c666b20d0f45e1fa3f2964b23f678cbc

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:440-443; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.leftCenterX

Source hash: 01fd3b14ca7e8a9d341e71e2fb86b773a9bea051016f7f6499489c3b65bab474

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:431-433; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.lowerChordTrace

Source hash: faa702d9afdff2eed044fad4474692be94c38bb3a9868779993853eb74445596

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:774-776; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.measurableSet_carrier

Source hash: c4a110b00efc328d9afc36bfb523f064a5cb69170025121ec86c0022fc798171

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:500-551; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.radius

Source hash: f23d45618cc8f0a402155f6b0b9c9bed86df54007d8b97c4f4027753ebfcd373

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:426-428; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.radius_pos

Source hash: 754f25da7343b067aca1ccf07fea956e09bfd461a16a072435b6b938ea98f462

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:460-462; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.rectangleCarrier

Source hash: 5204ba429abcba2c9a7bfd26b44272d1068d5b57b8a97a88331831661e9408f9

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:437-439; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.rightArcTrace

Source hash: 059982ba84229cdd58242cd7f98720be73ad274e207529f914e522d73ba5cbeb

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:767-770; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.rightCapCarrier

Source hash: 0f7231f5e97656f30e8e5131e2afc098206a3ff9953e2cd02ae8207ac4d54b12

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:444-448; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.rightCenterX

Source hash: ff1acc9e9b3ef170758d23460b0f614f6eb1c034ebd024b460f189e9f303f923

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:434-436; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.sideAngle

Source hash: 4f6b6599037149db7ce42f881c45d0eb7f35fbc43f124df7507d6a83a6d6e15b

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:429-430; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.sideAngle_le_pi_div_two

Source hash: 0c0857ab96c23693b4448efce2e70a6069dda6f460cf0118b7e365f0f1074987

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:466-469; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.sideAngle_lt_pi

Source hash: 4d165924e20daf34cd0cd3a93af12e6637e356347b723990beb9d1a85af59a9e

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:470-473; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.sideAngle_pos

Source hash: 20886da5ba412dbc0666239218b14d32a1bad68cf7a0f4837b3f6216a46aefbd

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:463-465; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.sin_sideAngle

Source hash: e85f9cb3c0c2b22c1ce74fdc60660033205e7a0821e76cfdd7248571352ea184

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:474-477; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.upperChordTrace

Source hash: 71db3f396f38ac0c4ed32b75273f08b0092359d4c0fdd7bc77976000e7791fb0

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:771-773; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.volume_carrier

Source hash: a426f0908aec7b73455f4a0093170232e841cf46ab8f580ee2fe442ff66a7997

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2202-2242; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.volume_leftCapCarrier

Source hash: ee7dbe308fa5b9feabca0b7d1f31d2c91cdf7e530a43dcc64a52dce7563fb7b8

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2076-2138; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.volume_rectangleCarrier

Source hash: b2c1d8716422a2e0169b6aaca489790c14b0594bc75f1e77b6bbec1349d5110

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2060-2075; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.volume_rightCapCarrier

Source hash: 7bc214af5d8d445665e416853a1fab80112ca5fe1116500de0e9b2b6285dfdf7

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2139-2201; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripCore.weightedArea_formula

Source hash: 22039501f3884648956188fd011ab629e4d948f733db9c19194b8b310134e20a

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2315-2333; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

StripDensity

Source hash: 51e629a8115629e4652f417b9dcfc46beea6a584d8ebcb351e746df181f97ab1

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:43-54; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly

Source hash: ba80fb031079587a301c08eeb69ef793b83ab9e2bcde30a061c829264b7e1959

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:29-35; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.HasModeledCompetitor

Source hash: 6f62c793e9d706d4618e398ebe39871915cbb898f90f26305f1f8dad0a496a66

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:3076-3080; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.ModeledCompetitorRealization

Source hash: 7461e543281be1bed3a7575ad4267ecf1122dd3b3edd989aadf4754226aea410

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:3043-3052; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.ModeledCompetitorRealization.asCompetitor

Source hash: 6c62dc13fbd8bcc95bc9815a8c9dace231a72bb0614d3ccc231b26790b97baab

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:3053-3054; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.ModeledCompetitorRealization.asCompetitor_weightedArea

Source hash: ac19f7d26c088671b4dd0552d598cd1124637e2815ea1238d0d5daf917f4ca9b

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:3059-3064; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.ModeledCompetitorRealization.asCompetitor_weightedPerimeter

Source hash: c73f289c45ecd5870b19789b66c484f95d8c8aad15bb42c8e172d87d107523d8

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:3065-3075; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.WeightedArea

Source hash: 5d054747c494e793da416a19e2d895420eb9e121366a5645e79b7ab53a0c3c7f

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:265-268; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.WeightedPerimeter

Source hash: ac2df4f2daf804e358c901d1a6b496eaadc35076745e389cd399733fc5dead10

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:269-272; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.angleTerm

Source hash: 8089feb5fd033f48e23e1ab4579cce22afbbbfd128400123f3b8b2b2a245332

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:289-291; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.bottomChord

Source hash: 08cc63d3aedc396b5efa9b27fd1bac931beb8b7d14cbce558e5fde942be277c2

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:136-139; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.bottomChord_eq_zero_iff

Source hash: 1b82dffbe772075bcb5095c9dc075644068e9232dc927ae0cd23919b863062dd

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:355-358; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.bottomChord_nonneg

Source hash: bbe5a0b68c27b1eab70889a99796728aff04d58c47eda0333c16191c78a8f12a

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:352-354; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.bottomChord_pos

Source hash: 5deb1da8ce95e59329d432d4f1484bad3b915566f05bd8a9e6b37a69ad4de177

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:359-362; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.bottomSegmentCarrier

Source hash: 18e70891a1386b33220d27cc96af45d0ffdcc88b970fd467c64035c04da589f5

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:132-135; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.bottomSegmentCarrier_subset_frontier

Source hash: 82d466e4484f4553cdf0fe29df6958b297f06ff314377fa09237775f676763e1

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1058-1079; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.boundaryJoinSet

Source hash: 4d107bd731d106706191e0cc66c1eb6e52b0ad83df797a9379d71e46e9ea68c7

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1130-1135; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.boundaryJoinSet_finite

Source hash: fb8b2d3307f8905b1616f7f8934468dae8dc2d1df098e8fa72f92f5a3759db5d

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1136-1141; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.boundaryTrace

Source hash: 77ded3af26dffe034b843162e20202cd8e868c69330a0cb95819f9e3b2a0bbf5

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1595-1599; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.boundaryTrace_subset_frontier_carrier

Source hash: daeda01e39ff76764527644f12850660eec76e1d4031dd8579c24a3b55dda6ba

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:2840-2847; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.carrier

Source hash: 0849c97e398a59f917657f18205fab9ffd6aa8e524ee751befd78f1c4735a4ff

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:262-264; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.continuous_realizedLeftArcParam

Source hash: 5aa13a8e0a54ece98427e269f77eefa81d89485ad9d1f32dde5c0e6f344ef9b4

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:690-699; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.continuous_realizedRightArcParam

Source hash: 3d9c0740b38c1fa009799b8c35785a0cb767211b94d4ca0bc37637a119a58afd

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:700-709; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.coreCarrier

Source hash: 5e82c9d38bf8a70ad9c2ae143c293bbf0238ee48148c3dd2d922c95471160878

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:99-103; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.coreCarrier_y_bounds

Source hash: a0a8c181bd45290d862033f8524f84e76d590a21a370fbbc0214aab783d9df85

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1447-1455; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.core_weightedArea_formula

Source hash: 5a80fac760bbad4e64452fa4d575b54c924b85ffbad6442cb8801547dabcbba7

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1530-1550; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.cos_innerAngle

Source hash: e774d0a2578a71402d098d5ac027f2d6d1dbb90425ca111832168cfb9f013412

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:178-183; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.cos_outerAngle

Source hash: 3eb066bbadef1d5d0722d2c322b7cfe8bce8e1eb62204f2872b1f8f1bbb41a24

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:184-188; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.cos_sideArcAngle

Source hash: a8f9a8c72dcee4cc4dc59b4da10f2ae6d2741d7865a4b2faca10168fa17cc76a

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:231-237; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.euclidean_frontier

Source hash: 5e03c2ded7f93ebce1b77460290c9d90f4e5be41990287351d94b69c624d60ea

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:2937-2949; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.exact_euclidean_bottomSegmentCarrier_hausdorffMeasure

Source hash: 56949577898a4e9f1638a1a8bab2c43a8263ea9e6cf0976fce47fbff56b3fe00

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1101-1129; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.exact_euclidean_leftArcParam_hausdorffMeasure

Source hash: c6dc496bb93e612df391bdcd13229ada6ddb03a05f7a4a85318498972631a8fc

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:742-778; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.exact_euclidean_leftArcTrace_hausdorffMeasure

Source hash: 9b484f3710815971495ad59b2a292d470a94ac2f3cad3edb96b8870979733b3

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:817-824; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.exact_euclidean_rightArcParam_hausdorffMeasure

Source hash: f1f573c6c931308699d4611129bc237b3c89fe21d3a82fbb640a2bc4f452f054

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:779-816; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.exact_euclidean_rightArcTrace_hausdorffMeasure

Source hash: 82deac992e3b3149a73b5a9867d35c880f6af8171bd8434cc5a883788eaac184

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:825-832; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.frontier_carrier

Source hash: 0ce2b73ef2aba31ce763d170404d962fb774d3c494bf5afe10757388adbfbdb30

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:2848-2854; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.frontier_carrier_subset_boundaryTrace

Source hash: e4a9f2900fa53b9204f0a28ca553872a35240978918b3cf32712160073f759cb

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:2443-2458; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.hasModeledCompetitor

Source hash: 6010bc1b92d07ee7af37566e925a969f227d520e7144371bb334d772b94b8386

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:3081-3100; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.innerAngle

Source hash: bb9acabc2bc9f5427d7041305c401b4e2793720759dd063e3153448db0793dd9

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:53-55; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.innerAngle_lt_pi

Source hash: 31385b2dc48409bb1698d05b2b8cedcea6151f9ce4bedc38534228b77f838b5c

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:168-170; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.innerAngle_pos

Source hash: e78599ffc8793999ca239c142df39a815df5b9220a19bd2bd882d169b85658b5

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:165-167; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.innerAngle_semicircular

Source hash: 136b95db3c fbe6ffba49a9d377637e6fd44e92cfd37653b368ff453e7d5af642

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:370-373; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.innerSin_pos

Source hash: 88694b992a38f1dffa3c67f32d082f4eb742c98f01874e62941887bfa3c13f04d

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:216-218; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.integrableOn_carrier

Source hash: f50bce5009ef96b4081db29960a7e72186fb76edbfedf9fa6fd538bc59ab8c22

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:3000-3011; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.integrable_frontierMeasure

Source hash: 242285e9e93302048576853114a4bc7f6ca16cdc917f22c33a2fe025def880a8

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:3012-3042; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.isClosed_bottomSegmentCarrier

Source hash: 4502cfa8cbb9cfeeb9921da9cbccfe0b4a2a7c969d9473aede4beb6ee9ada554

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1589-1594; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.isClosed_carrier

Source hash: 86d49c14fed38ff9d123ec7002dc853d7ab13d112817308f4d1ef5ac7d463980

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1307-1309; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.isClosed_coreCarrier

Source hash: c084bdc57b0172b2f03efa14200c62e0c7ef9b7e0265bac3ea60cec5c75fc390

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1303-1306; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.isClosed_leftSegmentCarrier

Source hash: 9a06aa24438ad23f7c7f01c4c87ea6fc01e48b3ecec4d358799e726bea34abb1

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1289-1295; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.isClosed_rectangleCarrier

Source hash: a040fbd7853bc5d56529ed0d45def53a17db2084e9e6f891db5067b23f32a560

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1282-1288; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.isClosed_rightSegmentCarrier

Source hash: 97ec89af687cf23451a62fdd3fb4816f90150533b2e6c5aca3fc7cb1fe8e300f

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1296-1302; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.isometry_sideArcComplexIsometryMap

Source hash: 5f9f98564eebaed09c3cd958295e743ceabd92f2b665793225e0fb64c4812ba4

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:653-664; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.leftArcParam

Source hash: 4f29c060ae66edf060cbebf1375e17d2584dc8c159343c7bce4a1ff009d7632c

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:120-126; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.leftArcParam_image_Icc

Source hash: a53d3bd58a5cf0ca117ca4045079faea921dc67e709030e1bc93b3bc04963479

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:446-546; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.leftArcTrace

Source hash: cfb74d8a54830ee4e695b37318a8d8c8a086063cc569f3e8d4fa167da3d5d0b9

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:104-110; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.leftCenter

Source hash: ae634544d50a625f2da0ab83dddef536c2004aefe8a002e19ec3ff792534bdf2

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:70-72; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.leftSegmentCarrier

Source hash: 7a6b1fb5a536fc1614b926138e279c79369045c2b292e00c145e1a0e699dbced

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:85-91; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.leftUpperJunction

Source hash: a660a79d351317f739116ce079a065c0c8bcd856fae9b73442ad8d1339dd7ec

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:946-957; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.measurableSet_carrier

Source hash: 25696aa660c5c71e8ce3e6a179a069d25633bc99ff664ad8e93ea6e319cb501f

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1310-1316; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.measurableSet_euclidean_bottomSegmentCarrier

Source hash: ac873340c9031353c9d26e2404b771c8a0ff1b638ea75b1a24ed627ed5bc802d

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:2855-2860; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.measurableSet_euclidean_leftArcTrace

Source hash: 791dc1e2fe6e2d0cb10c51a3bd4f4018c5ee53a25eeef24b22bc0d4e6f0f1192

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:732-736; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.measurableSet_euclidean_rightArcTrace

Source hash: 64197e0a340a1643ad8bc1525fe961f5b8792a0e7cb9321824fa2436e9d02bb4

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:737-741; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.modeledWeightedArea

Source hash: 35499fdca0a3bbbed11909d6dec0bbb1c83123d6ea5cb0599717ee289f567e87e

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:278-284; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.modeledWeightedArea_formula

Source hash: b0e70e8aeafad1e18dbaff1d3d6364f544c6042f48ce97bebd3fa5c6cc8e74e

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:885-895; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.modeledWeightedPerimeter

Source hash: 76198be6eb5b8790a4af34449bf29c4b963314e1e4bd268b4d3c9cdb2e69ea0d

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:285-288; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.modeledWeightedPerimeter_formula

Source hash: bb4c05a2ca2fd9d5b63be1b6ad16da716cc43497abcad228054765e6eaa2e39e

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:896-903; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.one_lt_radius

Source hash: 4e892eef63bc8252d5a243e2b5495e748b55fa956d516b3510c9e5289ec264d9

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:148-152; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerAngle

Source hash: 0e2ff2070394fdda5b4ec09824f47ed543dabdfef101c076ad378e0c92be2ff47

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:59-63; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerAngle_lt_pi

Source hash: ce135b6a3dc191fb33efd487aab9da31365b8ec038647a624b62a436ba911d34

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:174-177; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerAngle_pos

Source hash: 760f9b90fd549c278aa760178ec027d7c643f8b5bd6d4bf942c86f6a65bb0e28

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:171-173; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerAngle_semicircular

Source hash: 9ff26a1092e7089058ecf98f24fecce78972031b475870c0e4d30009e831f8ee

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:374-377; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerArgument

Source hash: d6cfc025aa1368c77ccefa6b050dbf8bfa113c35569b4434625ec2d735b91f41

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:56-58; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerArgument_mem

Source hash: fc2a8b1032448c01ece41eff0bcd5d748f70ee556d5721867bb36b59d56ce63a

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:156-164; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerCap

Source hash: 5482055613f2ec0454ddd9c71d722bea04494543bc659e447732870c662eb86f

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:251-261; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerCap_arcLength

Source hash: e4c6a09e492a4c9395262e1c277f2cd0b185e72ae34042a7f14138b12c6b4c19

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:875-884; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerCap_center

Source hash: 1057a7a30ad6e7558c1c0e586fc0147d20f1fc01e6c3e508acda901e55191c4c

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:929-933; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerCap_euclideanArea

Source hash: 3c7709918bd2e08d7c7ea431b339d946f2eb6c3c9d1138f4304b6afe64a153c2

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:851-864; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerCap_leftEndpoint

Source hash: 369e532ae6bbe742ee13471a1a8cc638a185e8c74c4f29cdde432417763b0fdb

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:934-938; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerCap_radius

Source hash: f107696210fe89f5a746d54584303c70da482f05587bc229f2d37aaa9494b57f

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:419-422; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerCap_rightEndpoint

Source hash: 0bf0b00b58f0396d9b9b8bc9e08617bcd76cf3f0b76937e23b52dec1f9b5fe12

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:939-945; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.outerSin_pos

Source hash: feedd827e6203dedcb21f4f474b9fea36ae3ea708b33b644b944093c5f958959

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:213-215; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.pairwiseOverlap_h1_null

Source hash: 545b8d3e4922237126e7364ee9bb07b444cf8cada72b02e8afca89501d2f79b3

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1258-1273; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.pairwiseOverlap_subset_boundaryJoinSet

Source hash: f8736f4dd5a67fc4149e15b1ed3e3f21030a49e12643cd517b41158566d4664f

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1142-1257; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.radius

Source hash: 44bdb144ee33a0fe634670e914d151b3f9a50f2b2f7db8da7c3fad23bdfbf70e

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:47-49; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.radius_mul_h

Source hash: e90d4095fef5e1e089527a179a53736b5dbf462b6149abecb5375e96b7380078

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:197-200; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.radius_pos

Source hash: a610fc617d9a92f1154e4b9627bb16a0d909464945f9e22ccb900c975acb2c7d

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:145-147; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.realizedLeftArcParam

Source hash: 51fbf9517aa10415de53dc106db635c5f70b9c39860ba75778e696cdeb2d6c3

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:665-668; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.realizedLeftArcParam_eq

Source hash: 538a7925bd9a6977d5e74933bd5f951885abb82dc534e3d102db01bf6aa01a1e

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:672-680; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.realizedLeftArcParam_image_Icc

Source hash: 06e6662a39cd836487840418c7224db9c1b8779c37aa295aab4e1109503a76c7

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:710-720; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.realizedRightArcParam

Source hash: 8a963395af5bf096a9c5748471f90364728d8fe511c4e2cc0591bd2068d9e016

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:669-671; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.realizedRightArcParam_eq

Source hash: 3339fa9296e8c62289cd021c3334658bb86b73401d06cffbaa790d7423ffd93e

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:681-689; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.realizedRightArcParam_image_Icc

Source hash: 9469dc298ecad88323016b9907111e322a864f0831a991b637ca11824881af16

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:721-731; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.rectangleArea

Source hash: 3314187b7b48ab958a7d8415bcf10afdb2dd85317d354dfd049f5344c2569e90

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:273-277; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.rectangleCarrier

Source hash: d451775e4018b5bae444c6a8ed6911c5adcc23db4a0efc68ab71c655f2933477

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:80-84; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.rightArcParam

Source hash: 74515bd4fc474dfa2c52a14662b8cc34aa0f40ef9d8185819e05d1de94ccf588

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:127-131; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.rightArcParam_image_Icc

Source hash: d47f88fb5eadc69e4f4bc7ce3bc201dc34d60ad570f34d3d29c4d0184ef0f8f1

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:547-648; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.rightArcTrace

Source hash: 828cebab0c56e67a2a3cfaae9e912a8a76f497287dcea6b589e1a22a40edd616

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:111-119; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.rightCenter

Source hash: c1d50740a1dfb76a9c424f19f6e78589f8a8ab930ab1727da5282aec42c3f25

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:73-75; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.rightSegmentCarrier

Source hash: 889596cedc4d7d957005581540478e3a458f9f3d1f37c9af6b1a713eee05b078

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:92-98; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.rightUpperJunction

Source hash: ae6c6a14cc5b76c0f3aa60155a1fb16c629e4d67ae173c7593aa69c135bbff39

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:958-968; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.scalarWeightedArea

Source hash: 195fdb382725116907cc721bab689c5c6f6f34b356a442f900ba649d89b62b9c

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:295-299; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.scalarWeightedPerimeter

Source hash: fcbac6602a2b133a8e714c37e1ef51bfb698ab0ae754dcd7eca0cb6e9c49204d

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:300-302; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.shapeParameter

Source hash: 29190b0ec0676db3ee8da889fc7b1e078bce75188f6b4271945e3fee045f2920

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:50-52; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.shapeParameter_mem

Source hash: 018fe2d07fb99b9420ec19a75327dc616f0119968b1c22fb83265de47b1261df

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:153-155; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideArcAngle

Source hash: c68849312591ff361810874588f817eab8327b61ad9b815244ad1ab61a55597a

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:64-65; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideArcAngle_lt_pi

Source hash: 27c4b01d9c8710ab675792367c9a16f3b7952720c452f471c97b46d487e08c08

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:223-226; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideArcAngle_pos

Source hash: fb3e4cdd6ff568be2285a238c6293705a5d18ec17ca0000cf9fc5fd3be6f53e2

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:219-222; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideArcComplexIsometryMap

Source hash: 2d2b0ae6442b1aeb7779e0e2c761b53071a3b1948418645b546ba443dcc3b322

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:649-652; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideCap

Source hash: 62a5fb2c6003ebc71ae1cea156d819929dd2061bdfd35d03ac33c3bf896e8fd0

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:238-250; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideCap_arcLength

Source hash: 0735084a84605f457abd29f126bfc3b41f13fe2e361355b34befa71a89121ae9

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:865-874; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideCap_center

Source hash: 5c32cc16475c4c5293c8fcbce19421c65fe608e9916172122a7cb5ff16d6b389

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:430-438; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideCap_euclideanArea

Source hash: b6b94a03ab653ee46e011e7fa757125fdbdfe26f1473cd6911d9100fe92583e4

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:833-850; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideCap_radius

Source hash: 860b391d38da56536b3934963761766a28238619240605477033823fc900d1af

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:423-429; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideHalfWidth

Source hash: c1a6374b950f6fdf60b58062fd5753e6cafd953a8fbac46cb6164191b66af1b3

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:66-69; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideHalfWidth_eq_zero_iff

Source hash: aa31042682af8c9fc65e3a19a4a9ce7dc71ed7e2ad3e807732dcfcc2da35644f

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:344-351; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideHalfWidth_nonneg

Source hash: be70cb71b25ed2f43754461915097af9183cc7aac3504655dda98d8a67b605ec

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:338-340; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sideHalfWidth_pos

Source hash: a769284707d6bf25198d747cd5846bae22f2f394cc9d09e601cfcab854579d7a

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:341-343; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sin_inner_le_sin_outer

Source hash: def468198882d033570072298a83ae1d41fc003f0ecb9d0b4ee3a6b94e6a1d10

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:303-311; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sin_inner_lt_sin_outer

Source hash: b3a6348544f17baf0cf9a8d51663f5b8b6187359294117163fc5d2afba78dd58

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:312-337; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sin_sideArcAngle

Source hash: 41165bd4a88985079154632e1304282b8868bb98cc6b1d97bb3ea19faaebe0dd

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:227-230; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.sineGap

Source hash: e9d7a7074e27fe28a23aa0db9c4cf9b0ef4e5f2181f1cfab6098f7cc2e240ecb

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:292-294; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.upperCenter

Source hash: 6dabb1e2d2dfef1af758b047dfb6728a97191f4c64742b75a35f102c81c1c3c1

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:76-79; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.volume_coreCarrier

Source hash: e422839d8a82e578aa8619bf8a6d240c5dc840fd4d4e1880ecc09a0307558380

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1481-1529; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.weightedArea_eq_modeledWeightedArea

Source hash: 5f7913f0171c133683a735330062b186a060d0e518987c5488549de7bfd06347

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1562-1583; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.weightedArea_formula

Source hash: 53616c2c25615a95257aa4773c87d4c69464dc905650e6ed5b53696dc5f08570

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:1584-1588; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.weightedPerimeter_eq_modeledWeightedPerimeter

Source hash: 3dd334c8f3cff9146f23a5d90e6da2520151c567d49c8604446b90aa7a9f4d2b

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:2950-2993; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

TypeThreeAssembly.weightedPerimeter_formula

Source hash: c6f51109b06f9f61cbbcb63e9d6e1d950e5a79c8decdf76b0e2fc3f366f8eca

Relation: equivalent

Checked against the exact frozen declaration at TypeThreeAssembly.lean:2994-2999; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

WeightedArea

Source hash: 959b5ddd7363861d4d0d79c211781b3b19e09df50087472690ae0581725b8c6d

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:55-60; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

WeightedPerimeter

Source hash: 8d1f1a60a6d4c5ce03ccda31c1bc34ec4f837b6ce55275e4b0c06fe04e566dba

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:67-73; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

area

Source hash: 8c44e8dff416b2208afe91aad54bf1678ad6f2299c75592fe1fed4f6cbf55dc2

Relation: equivalent

Checked against the exact frozen declaration at MorgansArcFunction.lean:92-93; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

area_pos

Source hash: 0af41b50988d0bace92fc2e638571302802d8538245a2c254bedb9f164463ede

Relation: equivalent

Checked against the exact frozen declaration at MorgansArcFunction.lean:148-157; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capBoundaryMeasure

Source hash: 5dbd12f9bc57860ad2ce92a638514e0e8d15006685aca7388e2fa2e1689504cd

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1249-1251; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capBoundaryMeasure_integrable

Source hash: 0065afd47de261467dd963dc3f68bd9fc5e9eab2c424cfff1d00f2e7c2919a7

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2260-2267; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capComplexIsometryMap

Source hash: 3ed78deaf1597a91c6218ca8fe10519ba693355e7329e0301f17fcce226f5a43

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:390-397; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capEnd

Source hash: a140ca726638df619aef507f8019eb9f7ca9503d3231f8bfa8f0a64feb377161

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1246-1248; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capParam

Source hash: d80d6cd5283176fe8b29770f039d1135a8892539144575a4bc1d9962ad427168

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1241-1243; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capParam_image_Icc_eq_arcTrace

Source hash: f412e9294ab946fde4ef494447fa75b0b57b0600c2bd61007639df1bb35c0e9e

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:947-1070; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capPartitionPoint

Source hash: 49ac6ecbe67b9040843eeee672d7b0f08055016781814c3ce0fa4e7d31ff805a

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:154-156; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capPartitionPoint_dist

Source hash: 6b90645e53e16e3855618b966b04af3ee0f15439043926ba1133c583e65b5a00

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:185-207; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capPartitionPoint_end

Source hash: b5d0e986af4c921461bd13ce89490a41fdc02991eaadda7d26f70db465413519

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:161-166; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capPartitionPoint_mem

Source hash: 8f26d09621c071ac8a6a1935a21253dfb29cb269e0ec21dc71e59a40f0529e9f

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:175-184; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capPartitionPoint_zero

Source hash: 1714782661300a21ee2d4243c7e4fbe79e1e97ce545ea187699c13d709db2db9

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:157-160; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

capStart

Source hash: 594eb6e2bbf3cbc7a2397b6af7a93506027168fa75ecf7f70b7012b2fc6e208a

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1244-1245; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

cap_arclength_eq_two_radius_theta

Source hash: edad6e24948a1499ce1a66a5efdc84d9c8f7899f5c52d6551716312e95163fb3

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:451-456; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

cap_integrableOn

Source hash: bbc4a1a9af0d45bee6aa1c1152d956e0e945a869f3550ca289ef7022556292ef

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2303-2308; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

cap_lower_external_density

Source hash: baaf83c56acf8edc18268269454b4946af9c7098831da004f1da856c3f80d9b2

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1341-1352; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

cap_lower_weightedArea

Source hash: af4bcab3c0cdacd24a9ba22f365ac164c96fd2be59faf0a944ca162de8e4ada2

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2360-2389; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

cap_lower_weightedIntegral

Source hash: 5575ee03ce1e095004c86962659c2bcf1dde6c0b1ad90b439220173f4e11e7fc

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1373-1394; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

cap_partition_polygon_lower

Source hash: 4dc7fcd936b6c56a4643f00c8d67c38ce24d6351f625a8873960fc070d61a446

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:208-268; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

cap_upper_external_density

Source hash: 0d4d779bde08a636e8536f594b78dfa9705a31925f9d2e9248d0d62fdf5aa11a

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1329-1340; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

cap_upper_weightedArea

Source hash: 2f2165b544886739ec40b854c29666e67f06c7a423bd046c04eb224a5c6ee0ce

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2334-2359; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

cap_upper_weightedIntegral

Source hash: 45644ff66dcb2be9e3cb6c626527848bf909f532df9be34ffa1977e24d25b8f4

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1353-1372; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

continuous_realizedCapParam

Source hash: 439602ca8a442d23dec8f858626a27ca71e9723ce90b04b56ee6e94cd89590f1

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:442-450; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

continuous_realizedLeftCoreParam

Source hash: 4dc1fbf3a43e8117f791c26e28bd5a2a2466716abc2324d51a2522841ccc46bf

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:727-735; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

continuous_realizedRightCoreParam

Source hash: db06971d2e0a6e54b3ce1b2cb1ffe296cb5ed7ac07495118f10c8f5f82bfe132

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:736-744; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

continuous_unitCircleArc

Source hash: 343bc4472a64687ca829ebbf897c060cb6c7c3cad3716259e96fb84cf091470c

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:14-17; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

coreArcEnd

Source hash: 7ee06e3535f2fd49529132649f006dc8125adff906e2bb1c8e325a0ca4b835a4

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1474-1476; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

coreArcStart

Source hash: 4c395d23f09d7608737696a9d4043f61933ddaffbb62953e4470c1085d950b4e

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1472-1473; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

core_integrableOn

Source hash: ebecbc727d05931a9afed5e77aa787703ada65169aa70e3f372bec6e04bddcac

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2309-2314; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

e11

Source hash: 9adb32c6bf9324acec1e791c0ec46f0e17df835ffee7b3549cab07bdc3a760c9

Relation: equivalent

Checked against the exact frozen declaration at MorgansArcFunction.lean:89-91; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

euclideanJoinSet

Source hash: 96d51d86f6c32a7f1c31ecb0290be17bec48e5d48a1aa38fdfbae4457025d389

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1257-1259; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

euclideanJoinSet_finite

Source hash: 9539ffd4336468eab518f7e951c9dc4db7ee2d6bd08f0fa98bbda9dcf0817f38

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1260-1262; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

euclideanPairwiseOverlap

Source hash: c1559aa1737fb404d99a3376b70fbd4f0ff514e71a947a7d30b01a1ce4c1177e

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1591-1597; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

euclideanStripDensity

Source hash: 3d074997fd7c2f4c47fc25c7e8a9a39259d6fd4628b103b9b3e4da77e1679848

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1389-1391; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

euclideanStripDensity_integrableOn_of_measure_ne_top

Source hash: 509b7e5c9b45ab61b5e958fce825f92bda813640616ddf5c7591ac420818c5d2

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1404-1411; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

exact_euclidean_capParam_hausdorffMeasure

Source hash: ce3a2c8ff8b380f7da5859563d117de2848052560e457ef5f88e25aa88808f43

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:457-483; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

exact_euclidean_capTrace_hausdorffMeasure

Source hash: 8a0602f46d010a0d38a019d5e1052cfe4c5d69fc3db5988eaac8c5dbcc5c5e1c

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1081-1088; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

exact_euclidean_leftArcTrace_hausdorffMeasure

Source hash: f6b756b038d2dd53ff9e214034534494d6216a6823376bb73c389a754a1ad0d9

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:821-827; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

exact_euclidean_leftCoreParam_hausdorffMeasure

Source hash: e0211c9e8b0941bc0bbeb3bd977afe28d78688025726aac98a4d738bf7f180e9

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:745-772; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

exact_euclidean_rightArcTrace_hausdorffMeasure

Source hash: 47e43c0cab0cb015904a2fe672eeec0720a25537d058b25dd2b767b05bcf1d39

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:828-834; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

exact_euclidean_rightCoreParam_hausdorffMeasure

Source hash: 5b9d17b9e5921b7fba027be4ebfa3845e68c3c43a92a67ce9d4854bf1b244db0

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:773-800; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

exact_euclidean_segmentCarrier_hausdorffMeasure

Source hash: f2e6ec79942c7df8c258fab2922138cb782506dbaa0184c81a94b12e6e9b9896

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:937-946; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

fourArc_euclidean_frontier

Source hash: 748ff0ec3e12377aa952eabaeafb59718df5bc069c8651bdcd45a9c687a1d41d

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1338-1348; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

fourArc_frontier_weightedPerimeter_eq

Source hash: e95ee1864c7bc7d9bc3a3ec2659625a510c6cefccdafe0cd2efab0e2d93e1067

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1794-1844; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

fourArc_pairwiseOverlap_h1_null

Source hash: 01eb58d468da4075ed5a4b3c4808bf8deddca98be0f05ef720c74c5a64fd9f4f

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1269-1279; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

fourArc_pairwiseOverlap_subset_joinSet

Source hash: eea91f9cc72f6aabc972107df3b4a4acd7547fe5fa392689f59302f202356a4

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1142-1195; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

hausdorffMeasure_image_Ico_eq_Icc

Source hash: 1c850c9b7ca6e169808ebfb87cac4902c44102258f4c3ba0c9b6433987f2a5de

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:142-153; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

hausdorffMeasure_one_euclideanJoinSet

Source hash: 7ebe8fae03ad531268ff769afa248cd1067ee674eba1a7f8bc77f36d3ab49ebc

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1263-1268; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

hausdorffMeasure_unitCircleArc_image_eq

Source hash: f4746d8c9b2c6ef1416bccca1f814590cdef08c393e10b7189ff167eb0402ab3

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:294-311; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

hausdorffMeasure_unitCircleArc_image_eq_of_span_lt_two_pi

Source hash: a5a68294ffad2a36064599c08079161b28ea03171c66437e00e8f24eba632c26

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:350-389; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

hausdorffMeasure_unitCircleArc_image_le

Source hash: 07ee598762aa4a7964c4db549c22833ebf32ee2d325dcc190dd9a72212015319

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:93-104; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

image_Ico_eq_image_Icc_diff_endpoint

Source hash: c4fb0e64c0405d8f9fa3cb2429d9a9bc01ac95be8aba1e08560d1aeae3f939e9

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:124-141; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

integral_four_union

Source hash: 8b3fbc02eae74d84636263a1b0bedcfff91c58bcc3410ac92807b51d54cc9ea4

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1562-1590; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

integral_semicircle_scaled

Source hash: fe66bbeed922271b7cf0e4043f58e6b4f6651310c0d02aa776383ac42b18240d

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1759-1791; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

integral_semicircle_translated

Source hash: 06721e5dc911d180205234ff69b820e369a8d82b66f142e17c0a1377b9b27f8d

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1792-1800; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

integral_sqrt_one_sub_sq_from_cos

Source hash: e4a969c6c1b67c6f267db8fb9582c7cd96174fa24489265c840c646c3085a52a

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1729-1758; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

intervalPushforward

Source hash: 366d70f2307ddf912f8fe8519c486284fdacdab2c9bb76f371f9a65dd8ee807d

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1200-1202; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

isometry_capComplexIsometryMap

Source hash: 5d4136606591a26f6abf24f8ba286ddba7154e77e9f96c81673196c1854009b0

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:398-408; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

isometry_leftCoreComplexIsometryMap

Source hash: a61c4ab4c222e88c43376bf46386ccf7697c506b7a23f81458ff102288a9e14a

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:685-695; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

isometry_rightCoreComplexIsometryMap

Source hash: 2857926ac4225e60a51ede217224fdFDA63879d202e0be14411ea5c13b2fc6bf

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:696-704; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

isometry_unitCircleArcRotation

Source hash: 0390b164e94132c53e908a7960b31f68dc786e7272a9a9f6482fa79cade88362

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:315-322; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

joinSet

Source hash: 613dcfda543c5255d80175237d7db38b3386ce55bc3215682252569ce26c42d5

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1115-1117; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

joinSet_finite

Source hash: c95df1795c640672a88890bc658373f829154b7fa3a2361c967c14e17f22b551

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1253-1256; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

leftCoreBoundaryMeasure

Source hash: e243bfd8c440c79be3e701f3e1ba7f0b4af716d3ad6d1c3de2eb88023b1a30c8

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1487-1490; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

leftCoreBoundaryMeasure_integrable

Source hash: efdd79d96b8f43a7c478608fb7242070e102fd586af27d2826a566f07542e0a9

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2275-2281; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

leftCoreComplexIsometryMap

Source hash: c0385e604fc62ea1f5b752f17a4538ba2edeedf20e52cbfb6848b3b966b2ab57

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:677-680; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

leftCoreParam

Source hash: 80f940fe52c510e69a283605275f10b49a7ca369836a80a40ece896dbbf606db

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1477-1481; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

leftCoreParam_image_Icc

Source hash: 01c2af95d70e04d0bac530bc681fda34196e33e841f34e26466b4a3a9c6ecd18

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:517-596; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

leftCore_strip_density

Source hash: 3222d64a655ce3af15a638606821b923e89c31ae75558031c08763aeb76b72b6

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1662-1667; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

leftCore_weightedIntegral

Source hash: e333f475432d30c40456ab56a9f5187cebdb61288c7e07d5eec293804ac77f49

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1675-1694; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

lintegral_semicircle_translated

Source hash: 3c8e90cd635d133e832b64fe2731d46db4f68d0a1a341d292df3904cf66e67e8

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1801-1819; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

lipschitzWith_unitCircleArc

Source hash: 6ddc026dd017d7b20d15e385a48bc34f5bcbfcad3796885aedaeb1fd06730313

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:33-54; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

measurableSet_euclidean_capTrace

Source hash: 345ff49d99187e89c9b6a7558ffa41e164108dfd32b3dc525b699de014a17b5a

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1104-1109; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

measurableSet_euclidean_leftArcTrace

Source hash: a31ed55013e95ae690ffcc5bab6587c2933302f4970b8b673a167e6c53e1b7f8

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1089-1093; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

measurableSet_euclidean_rightArcTrace

Source hash: 771198ca90f5054f065ccd16a46e41b5fce0886173f7b3c654e6b61824be845e

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1094-1098; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

measurable_stripDensity

Source hash: b6fc908d8bba478440a5c35f7c75a94366dc691bdf7d2125848e2a8f512284af

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1234-1240; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

monotone_capPartitionPoint

Source hash: 62e77199a2eb7f24e5f93307536675c49f368eab2d84be96574f544f2152a857

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:167-174; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

ofReal_dist_le_hausdorffMeasure_image_Icc

Source hash: 272ec4c298bc0f3582f8dd04d91ee8113b10ec4ec1c37538e792ec5391cc0936

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:105-123; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

pairwiseOverlap

Source hash: f2051dcd764bb55c1f4c980a58c56ba3f51d928d6141df3262dbdd919f1b5fbd

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1110-1114; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

pairwise_aedisjoint_of_overlap_null

Source hash: 4b46ea8bfc070cb38c17e38f03c2486a1eb119281a4706f7a55c5da843e21d49

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1598-1616; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

planeEuclideanHomeomorph

Source hash: 0cae97f48ab4c061d7f08a907381bf52e9560477df33ffb993c23c2cbbf11049

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:38-42; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedCapParam

Source hash: 95b1661a30d7b5d3f17854c6eabbe1a4da9d944d9ad542e596607c6cb4fd12b0

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:430-433; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedCapParam_eq

Source hash: 9c421cce21d93c49e9d419ecc30c8d31de995d391da25f034d29cf3f0bd85599

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:434-441; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedCapParam_image_Icc

Source hash: a37b40496e1e5c2ae0b778fe01e7fee7a91557ce9473bf6e0b7755afebbd941b

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1071-1080; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedLeftCoreParam

Source hash: 854e6d1ea17e2b2d6576f273610fd43051ac61c0bb7e75c185b3d3aa821f7da4

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:705-708; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedLeftCoreParam_eq

Source hash: e2c0026193d6fa0dd58c252b75c6df1a108616049c60130059ddb93311c0b2a0

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:713-719; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedLeftCoreParam_image_Icc

Source hash: 43b7188db3ec73adee2747ff0dc65e5368084e5e17965fa7fdaa8f3c9301ccd7

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:801-810; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedRightCoreParam

Source hash: 4f8de565538cdbe799e2faf0a2d6452b4030fcfacd5b43a7347015b8e46924a9

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:709-712; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedRightCoreParam_eq

Source hash: 4aab5f14bc5465771d4021be2ff0ba59e74b69eeaba0b9a9c834bd0bf72ed625

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:720-726; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedRightCoreParam_image_Icc

Source hash: e28432904f0d4f1c849ed6ed56e30bdae00185a8e05f01ccceab8ef89d16f29b

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:811-820; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedSegmentParam

Source hash: 4c1728c18c2df9a1da871e357fce4f6439c6307fe654c71c273fe06b8caaa0fd

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:864-867; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedSegmentParam_image_affineSegment

Source hash: c3be9a5bc2d447d5c5665690a46411881b4f0933eb5c3cf50deaa41a88946c72

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:873-912; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

realizedSegmentParam_image_carrier

Source hash: fc21a65690cb2d00a27cb6aebde20046f1dffbc15b7f30d3caf58ac1aefd1725

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:913-922; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

rightCoreBoundaryMeasure

Source hash: 33cbc13fa6fcd71207ac7e5dfb0579966744c8887bb1c9af1b95e6c688f55ea8

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1491-1493; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

rightCoreBoundaryMeasure_integrable

Source hash: 63cd5ea28e72c7cf39aef8bfd2e698b16940d7ae74acc20f074d60c55994913c

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2282-2288; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

rightCoreComplexIsometryMap

Source hash: 24206b125ec8e6de86a054126e34b32568c5cb9c365728de42d394ff9cc0ede6

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:681-684; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

rightCoreParam

Source hash: 38ea0d845e704b5d320adc3936a1b6a5fba52e17ce5fb320da7c20200afa9e4d

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1482-1486; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

rightCoreParam_image_Icc

Source hash: 78bb53ade5c270538e82ae914ea314e16857498de0ee3dfc73dde7468e65c773

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:597-676; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

rightCore_strip_density

Source hash: d879e18591ac7ff00f0a228fad33d85d1e290fb3c77820aaf80c672ec9411662

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1668-1674; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

rightCore_weightedIntegral

Source hash: 12c65212e5417b38acb8560f789f61c31ede351c3de6efe2b847438f7d6cf718

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1695-1714; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

segmentEnd

Source hash: 161abc8655ce59212c511d028e178416823adf64c523c73dae6b889ae03ed1f4

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1400-1402; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

segmentEndpoint_edist

Source hash: 667e4fe55a19b902ccd7cc29782bd67f737d862a1bddf625b46565ed0a83a1cd

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:923-936; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

segmentParam

Source hash: ba8ed6bbc5cd19b9c405036bea7c221ae24e5c76ea0f6ef5cc2d62c327ae1dc5

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1395-1397; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

segmentParam_image_Icc

Source hash: 78cfdac8328c8d0308b83fec45a7d72c3a0a830edb6a3f41f9bd2da54db0f76

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:835-863; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

segmentStart

Source hash: e90adf884523aab9d2e3af449adbc9d655d375051a612c0b4970a212a5dca61f

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1398-1399; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

stripDensity_integrable0n_of_volume_ne_top

Source hash: d26957a576fdcc692079b92ac5054cbbeda2a914af15d77e6d70ad1d418d20fd

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2289-2302; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

stripDensity_integrable_intervalPushforward

Source hash: 6315c9471ea2031d23e0cb4552b23f2812fe90938358c371a246f04d7f2d8095

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2243-2259; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

tendsto_cap_polygon_length

Source hash: 32d7803e3d290677cdac6cd7bd331241d0a1e99888d3a8665e326de2ca241a11

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:269-293; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

unitCircleArc

Source hash: 6865b0b68c07403ee39252efa35feb9b50850a4644c2bf915fb506723c7a07c7

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:11-13; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

unitCircleArcRotation

Source hash: cfdc45400ed0d08a69b42a215854d1f81999a500771f48a7f18476aafefe48ef

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:312-314; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

unitCircleArcRotation_apply

Source hash: dcedc37c9428c0a722c5bf130023e2389a83c7faa4036432847247c3401f6b9b

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:323-331; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

unitCircleArcRotation_image_Icc

Source hash: 6bbfe5bd6af2f20e278e2e30ff5adc9fe7c0761359113d169ea5baae1f20d6a2

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:332-349; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

unitCircleArc_dist

Source hash: 1c583965a8663dcd3d0483716b4801199d867b2ff4cbe96f04d4cbeb57ff9c08

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:55-66; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

unitCircleArc_eq

Source hash: 9342d6a8ed8b7dcd1fbce36251ddf6b271aeea6ec0298f5d467e178c8fbf9cec

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:409-415; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

unitCircleArc_im

Source hash: 0666e8d860656cd1bc24ade42af99acdafd9034497a0ba0f27acf44cddcd77e5

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:423-429; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

unitCircleArc_injOn

Source hash: ceabbbb1bd00ed7b3d5a248b0d9d8b4d41f8f1fc0af8194bd56a8092deed637e9

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:67-92; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

unitCircleArc_re

Source hash: 238aa74976048276575d89eb69760ceb968a224b5358a1775e61818ec60df916

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:416-422; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

unitCircleArc_sub_factor

Source hash: d6c59e7f498e1352e21969a1c8084fa6600038a792db1172f4e5bd3b2836eece

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:18-32; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

upper_cap_integral

Source hash: 447beffa9af9c0047e982aa709b3b243852770c1fde70cc3c6676215458bf

Relation: equivalent

Checked against the exact frozen declaration at FrontierPerimeter.lean:1545-1552; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

volume_horizontalLine

Source hash: ea745db5e4b12e1d05fe0934ca3c81eb24b84ea87dd522e3aa6824c86d1a2dce

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2029-2036; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

volume_lower_circular_segment

Source hash: 2301e25d7ced8d1d9e2c23a9b63e02ba7c6e8f9ea615112d4c41d15aeb777045

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1945-1995; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

volume_upper_circular_segment

Source hash: d23f6efcc432aaa58959c64bfa3f6f589362c33546093657b1717ec5dbfb1b34

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:1820-1944; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

volume_verticalLine

Source hash: b32e9d96eca7ded1e8b5354b6d117e61bcda7b34c1fe056587cd14e1871da61b

Relation: equivalent

Checked against the exact frozen declaration at CMVGeometry.lean:2037-2046; it preserves the handoff statement's parameters, hypotheses, quantifiers, domain, boundary cases, symbols, and conclusion.

Findings: none.

Generated declaration audit ledger